



OPEN

DATA DESCRIPTOR

A longitudinal dataset of physiological, hormonal, metabolic, and self-reported menstrual health data

Georgianna Lin , Jin Yi Li, Kaavya Kalani, Khai N. Truong & Alex Mariakakis

The mcPHASES (menstrual cycle Physiological, Hormonal, and Self-Reported Events and Symptoms) dataset provides a multimodal record of menstrual health that integrates physiological monitoring, hormone measurements, and self-reported experiences. Forty-two Canadian young adults who menstruate participated in a 3-month observation period, and 20 of them also completed a second 3-month observation period. During data collection, participants wore Fitbit Sense smartwatches to capture diverse physiological signals and Dexcom G6 continuous glucose monitors for metabolic data. Hormone levels were obtained using at-home Mira Plus urinalysis tests, and daily symptom and lifestyle information (e.g., pain, sleep, stress) was reported through surveys. In total, the dataset comprises 23 structured tables organized by signal category, allowing for analyses that link endocrine dynamics to wearable-derived measures and self-reported outcomes. This resource supports investigations into cycle variability, hormone-physiology interactions, and contextual influences on menstrual health, while also offering benchmark data for developing predictive algorithms and advancing menstrual health informatics.

Background & Summary

The menstrual cycle reflects a complex orchestration of endocrine, metabolic, and physiological processes^{1,2}. These processes influence and are influenced by factors ranging from stress and physical activity to nutrition and sleep, leading to substantial variability both between individuals and within the same individual across cycles^{3–5}. Given its tight coupling with various bodily functions, researchers have often advocated for the menstrual cycle to be considered as a “fifth vital sign”, arguing that changes in cycle regularity or symptom severity can serve as early indicators of underlying health disruptions^{6,7}.

Despite its clinical and physiological relevance, menstrual health remains disproportionately understudied. The study of menstruating bodies has been historically overlooked and marginalized in medical research, leading to critical gaps in knowledge about menstrual physiology across varying routines, ethnicities, and environments^{8–10}. These gaps are further compounded by the scarcity of accessible, high-resolution datasets that integrate hormonal, physiological, and behavioral data over time. While consumer wearables now offer the ability to monitor variables such as heart rate variability, skin temperature, and sleep quality in real-world settings, most studies leveraging these tools rely on proprietary datasets without access to ground-truth hormone data^{11–14}. Conversely, studies that include direct hormone measurements are rarely conducted in ambulatory or longitudinal settings, precluding investigations that require data across an individual’s menstrual cycle^{15–17}.

In direct response to long-standing calls for more inclusive and high-resolution datasets related to menstrual health^{8,10}, we have curated a repository called mcPHASES: **m**enstrual cycle **P**hysiological, **H**ormonal, **A**nd **S**elf-Reported **E**vents and **S**ymptoms. mcPHASES was collected from 42 Canadian young adults who menstruate. Participants wore Fitbit Sense smartwatches and Dexcom G6 continuous glucose monitors (CGMs) to capture real-time physiological data and completed daily diaries to report symptoms such as cramps, stress, and sleep quality. Using Mira Plus Starter Kits, they also completed daily urinalysis tests to measure their hormone levels. All participants completed this protocol for a 3-month interval, and twenty of them repeated a similar protocol for a second 3-month interval years later to capture greater intrapersonal variability.

University of Toronto, Department of Computer Science, Toronto, Canada. ✉e-mail: blue.lin@mail.utoronto.ca

	Interval 1 January–April 2022	Interval 2 July–October 2024
Number of participants	42	20
Age (yrs, mean ± std)	20.9 ± 2.8	22.0 ± 2.3
Self-Identified Gender (N, %)		
Woman	37 (88.0%)	18 (90.0%)
Gender-fluid	1 (2.4%)	0 (0%)
Non-binary	2 (4.8%)	0 (0%)
Questioning	1 (2.4%)	1 (5.0%)
Unanswered	1 (2.4%)	1 (5.0%)
Ethnicity (N, %)		
African	1 (2.4%)	0 (0%)
Caucasian	9 (21.4%)	5 (25.0%)
Indo-Caribbean	1 (2.4%)	0 (0%)
Latina	1 (2.4%)	1 (5.0%)
Middle Eastern	5 (11.9%)	2 (10.0%)
East Asian	14 (33.3%)	6 (30.0%)
Southeast Asian	11 (26.2%)	6 (30.0%)
Self-Reported Frequency of Menstruation (N, %)		
Regular	29 (69.0%)	13 (65.0%)
Irregular	13 (31.0%)	7 (35.0%)

Table 1. The demographic breakdown of our study population during both data collection intervals.

The mcPHASES dataset offers a unique opportunity for researchers to explore the interactions between hormonal states, physiological responses, and lifestyle factors in the context of menstrual health. The hormone data can be leveraged as both a physiological signal and a means of identifying phases of the menstrual cycle. This affordance has the potential to enable the design of menstrual cycle prediction algorithms that are more accurate and robust. More importantly, the mcPHASES dataset is intended to support investigations that appreciate the menstrual cycle as a critical window into systemic health rather than just a reproductive phenomenon.

Methods

In this section, we describe our participant recruitment and data collection protocols. We also describe how we processed the collected data to produce the final dataset. This study protocol was approved by the Research Ethics Board at the University of Toronto (Protocol #41568), which evaluates research according to institutional, provincial, and federal practices along with the Declaration of Helsinki to ensure best practices in the protection of human subjects. All participants provided written, informed consent to participating in the study via electronic signature. Consent materials described the study procedures, potential risks and benefits, data handling practices, and participants’ right to withdraw at any time without penalty. Data may be reused for future research projects that are consistent with the mcPHASES data use agreement, ensuring that any secondary analyses adhere to the protections established in this study.

Participants. We recruited 50 participants by advertising the study in social media groups and workspaces operated by women’s health advocacy organizations in the Greater Toronto Area. Participants were required to be menstruating individuals over the age of 18 who did not intend to travel outside of the Greater Toronto Area for significant durations of the study. Individuals who had diabetes were excluded from participating, as were those who had used hormonal therapy or contraception at most three months prior to the study. Participants were told that they could withdraw from the study at any point of their own volition. Those who became pregnant or failed to adhere to required protocols were asked to de-enroll; however, no participants fit this criterion in our study.

We conducted two rounds of data collection — one in 2022 and another in 2024 — with the intent of engaging the same participants. In addition to initial study participation consent, all participants went through a separate informed consent process at the end of the study to explicitly authorize the inclusion, public release, and future use of their de-identified data as part of the mcPHASES dataset. Data sharing was optional and not required for study participation. Table 1 provides the demographic information of those who agreed to be a part of our publicly released dataset. Of the 50 participants who enrolled in the study, 42 from Interval 1 consented to have their data released, including all 20 who rejoined and consented after Interval 2.

Data Collection and Extraction Methods. To select the types of data that would be useful for menstrual health tracking, we first reviewed clinical literature to identify health indicators that have been linked with the menstrual cycle. We then reconciled this list with commercially available tracking devices to identify suitable data collection approaches for our research. Given the commercial nature of these devices, they may have leveraged proprietary algorithms to process some of the collected data. We acknowledge potential limitations associated with algorithmic biases and, to the best of our ability, present the data in its most unprocessed form.

We created new accounts on all of the devices using unique alphanumeric identifiers so that participants could collect data anonymously. Updates to the tracking devices across the two study intervals resulted in the

addition, removal, or modification of some measured variables. The specific devices relevant to our dataset are described below:

Self-Report Diary. Emulating the standard approach to menstrual tracking¹⁸, we created a basic smartphone app that allowed participants to self-report the timing of their menstruation. The app also inquired about the intensity of participants' physical activity (5-point scale from "very low" to "very high") as well as stress and sleep issues (6-point scale from "did not experience" to "very high"). Participants were asked to complete the diary on a daily basis.

Hormone Analyzer. The Mira Plus Starter Kit (<https://usd.miracare.com/products/fertility-plus-starter-kit>) measured luteinizing hormone (LH) levels, as well as urinary metabolites of estrogen (estrone-3-glucuronide, E3G) and progesterone (pregnanediol glucuronide, PdG). Participants were asked to use the kit daily, after which the results were automatically uploaded via Bluetooth to the Mira smartphone app. Since we were unable to extract data from this app, we asked participants to manually enter their measurements into the smartphone app we developed for the study.

Because it is a cutting-edge technology, the accuracy and reliability of Mira's device have been examined across several studies. Nakhuda *et al.*¹⁹ established a robust correlation between urinary E3G concentrations measured using Mira and serum estradiol (E2) concentrations determined by immunoassay analyzers. Others have reported that Mira has 99% accuracy at ovulation prediction and a coefficient of variation of 20%^{20–22}.

Fitness Tracker. Participants were instructed to wear a Fitbit Sense smartwatch (<https://www.fitbit.com/global/us/products/smartwatches/sense>) at all times to measure several vital signs (e.g., heart rate, body temperature) and physiological metrics (e.g., physical activity, sleep quality). The smartwatch automatically and continuously uploaded data via Bluetooth to the Fitbit portal. Since we created new accounts for each participant, we were able to manually perform a one-time extraction to pull all collected smartwatch data at once.

Continuous Glucose Monitor. Participants were instructed to wear a Dexcom G6 continuous glucose monitor (CGM) (<https://www.dexcom.com/g6-cgm-system>) at all times to measure their blood glucose levels. As with the Fitbit, we performed a one-time extraction to pull all glucose data that was automatically streamed to the Dexcom portal.

Study Design. *Interval 1: Multimodal Menstrual Health Data Collection.* Between January and April 2022, participants used hormone analyzers, fitness trackers, CGMs, and self-report diary surveys to capture various aspects of their menstrual health. Participants were given financial compensation biweekly to encourage study adherence. They received \$5 CAD for each day of complete data collection, which entailed completing the manual data collection procedures and wearing both the fitness tracker and CGM for at least 18 hours.

Interval 2: Wearable Data and Hormone Data Collection. After examining the data accumulated from the initial effort of our study, we recognized the benefits of having additional longitudinal data from the same individuals to account for gradual shifts that may have occurred due to changes in their routines and environments. We invited participants from the initial effort of data collection to complete a second round between July and October 2024, 20 of whom agreed.

Since participants from Interval 1 often commented on the burdens of using multiple devices to track their health, we limited this round of data collection to hormone and fitness tracker data. In other words, we did not ask participants to track their glucose levels, and the self-report diary surveys were made optional; participants still used the app to manually copy over hormone levels reported by Mira. To further improve adherence, we increased the biweekly compensation amount to \$8 CAD for each day of complete data collection, which required completing the hormone tests and wearing the fitness tracker for at least 18 hours.

Data Processing Procedures. In addition to providing participants with accounts that had unique alphanumeric identifiers, we made an additional pass through all collected data to remove any personally-identifiable information to minimize risk of re-identification. We adopted the de-identification standards outlined in the Health Insurance Portability and Accountability Act (HIPAA) Privacy Rule, removing elements such as names, email addresses, phone numbers, device identifiers, and serial numbers. We also removed all calendar dates, instead representing time as relative "day-in-study" values (i.e., the first day of a participant's enrollment is marked as Day 1). To further support potential analyses of temporal patterns, we added a column indicating whether each entry occurred on a weekday or weekend.

Mira estimates each user's "fertile window" and menstruation days using a proprietary algorithm that identifies rises and falls in the collected hormone data. We used this information to label four phases of the menstrual cycle: late-follicular (the last day of menstrual flow to the first day of the fertile window), ovulation (the fertile window), luteal (the last day of the fertile window to the first day of menstrual flow), and menstruation (days with menstrual flow).

To present the data in its most unprocessed form, we did not exclude missing data or perform processing steps beyond the aforementioned phase labeling. We believe that this approach empowers future studies to explore and experiment with different data-handling methods. Since the signals collected in Interval 2 were generally a subset of those from Interval 1, we stacked the data from both intervals together to avoid having duplicate tables with overlapping columns. Entries were left blank for measures collected only in one interval but not in the other.

Data Source	Table	Description
Pre-Study Survey	Demographic	Contains baseline participant information such as year of birth, gender, education level, and age at menarche.
Self-Report and Mira	Self-Report and Hormone	Includes daily self-reported symptoms tied to the menstrual cycle like mood swings, cramps, and menstrual flow, as well as LH, E3G, and PdG hormone levels.
Dexcom	Glucose	Contains continuous glucose monitoring data such as glucose value, rate of change, and event subtype.
Fitbit	Active Minutes	Tracks activity intensity levels throughout the day, including sedentary, lightly active, moderately active, and very active minutes.
	Active Zone Minutes	Measures time spent in various heart rate zones, including total minutes in each zone using heart zone IDs.
	Altitude	Captures altitude measurements over time, recorded with timestamps and elevation values.
	Calories	Logs the number of calories burned with corresponding timestamps.
	Computed Temperature	Provides temperature data during sleep, including nightly averages, standard deviations, and sleep window timestamps.
	Demographic VO ₂ Max	Estimates cardiovascular fitness using VO ₂ max values, VO ₂ max error, and their filtered versions.
	Distance	Records distance traveled over time with associated timestamps.
	Estimated Oxygen Variation	Captures variation in blood oxygen levels using the infrared-to-red signal ratio.
	Exercise	Includes logged activities with metrics like activity name, duration, calories burned, and average heart rate.
	Heart Rate	Records overall heart rate with timestamps and confidence level.
	Heart Rate Variability Details	Details heart rate variability using measures like RMSSD, low frequency, and high frequency power.
	Height and Weight	Lists participant height and weight measurements at the beginning of Intervals 1 and 2.
	Respiratory Rate Summary	Summarizes breathing rates across sleep stages with data such as REM breathing rate and standard deviation.
	Resting Heart Rate	Captures resting heart rate values along with date, time, and associated error estimates.
	Sleep	Tracks sleep stages and timing with details like start and end time, duration, minutes asleep, and time in bed.
	Sleep Score	Provides a nightly score reflecting sleep quality, including components like revitalization score and resting heart rate.
	Steps	Logs step counts with timestamps for each recorded entry.
	Stress Score	Reflects daily stress levels using a composite of sleep, responsiveness, and exertion metrics.
	Time in Heart Rate Zones	Tracks minutes spent in different heart rate zones as defined by Fitbit.
	Wrist Temperature	Records nightly wrist temperature deviations from baseline with timestamps.

Table 2. Details about the tables composing the mcPHASES dataset.

We did not perform any additional processing beyond what has already been described to preserve the authenticity of the data and allow for diverse analyses. However, it is important to note that internal firmware and software updates by the device manufacturers may have resulted in discrepancies across the two study intervals with respect to measurement accuracy and precision; this is particularly relevant for algorithmically-derived measures like sleep duration and menstruation onset.

Data Record

The dataset repository for mcPHASES is publicly available on PhysioNet²³ at <https://physionet.org/content/mcphases/1.0.0/24>. The dataset comprises multiple tables linked by the participant identifier `id` and the study day label `day_in_study`. For measurements that may span two calendar days, additional day-level labels are included alongside `day_in_study`. For example, sleep data includes both `sleep_start_day_in_study` and `sleep_end_day_in_study` to capture the full duration of the event. Each participant's Day 1 corresponds to their first day of Interval 1 in the study. If a participant returned for Interval 2, their timeline resumed around Day 905, reflecting the number of days elapsed since their initial enrollment.

Data Structure Organization. The repository comprises 23 tables that are enumerated in Table 2. The first table contains demographic information that remained unchanged throughout the study. The second table contains daily information that was collected through our self-report diary, including but not limited to self-reported menstruation dates and hormone levels transferred from Mira. The remaining tables contain high-frequency sensor data from the Dexcom CGM and Fitbit devices. Each table corresponds to a small handful of closely related measures, largely in their original form from the raw data extract.

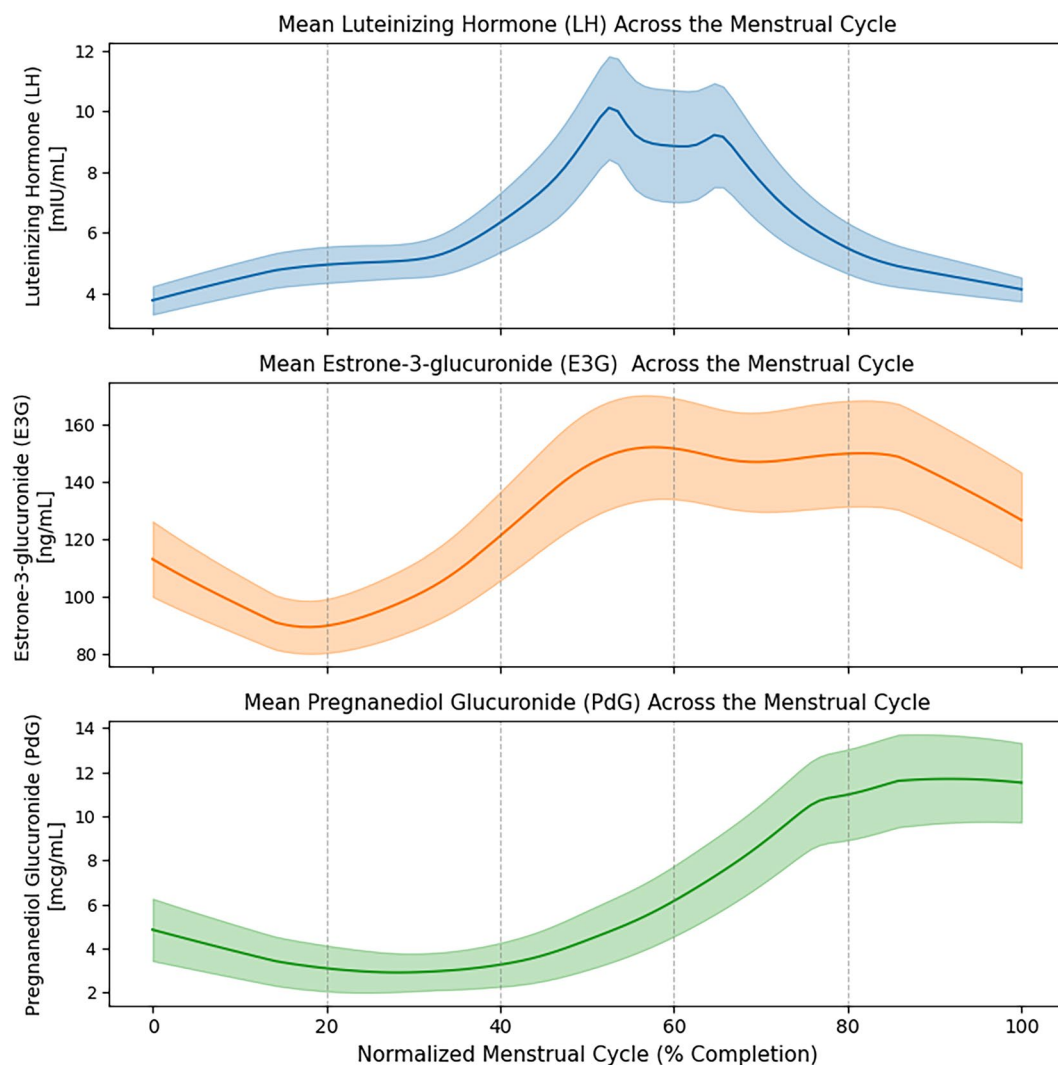


Fig. 1 A plot of daily luteinizing hormone (LH), estrogen (E3G), and progesterone (PdG) levels across the menstrual cycle. The horizontal axis is normalized according to menstrual cycle length and begins with menstruation onset. The shaded regions indicate 95% confidence intervals.

Technical Validation

Data Quality Control. Participants received clear instructions on best practices for using each device to maximize data integrity. Since the Mira Plus Starter Kit yields its most accurate results when users abstain from liquid consumption for at least two hours prior to sample collection, participants were instructed to perform the test shortly after waking each morning. Participants were advised to charge the Fitbit daily for brief periods (e.g., during showers) to minimize data loss due to battery depletion. Likewise, participants using the Dexcom CGM were instructed to replace the sensor approximately every 10 days as recommended by the manufacturer for continued functionality.

Participants were asked to report whether they had completed data collection using the Mira, Fitbit, and Dexcom devices each day as applicable. We compared these responses against visualizations of data completeness that we generated in a custom dashboard. Every two weeks, we conducted manual audits for missing data and reached out to participants via their preferred communication method (e.g., email, Discord) to improve study compliance. We also used these periodic check-ins to verify self-reported hormone levels against values recorded in the Mira app installed on participants' smartphones.

Menstrual Cycle Characteristics. A total of 192 complete menstrual cycles were recorded across both data collection intervals; this included 135 cycles from Interval 1 and 57 cycles from Interval 2. A complete menstrual cycle was defined as a consecutive sequence of all four phases. On average, menstruation lasted 6.2 days, the late-follicular phase lasted 7.7 days, the ovulation phase lasted 7.3 days, and the luteal phase lasted 10.4 days. Since menstrual cycle length can vary across individuals and within the same individual, we compare data across cycles by assigning each day a percentage representing progression through the current cycle, with the onset of menstruation defining the start and end. Phase transitions occurred with varying spacing across cycles, so we do not assign them with specific percentages; however, the transitions generally occurred at 19.6% (menstruation to

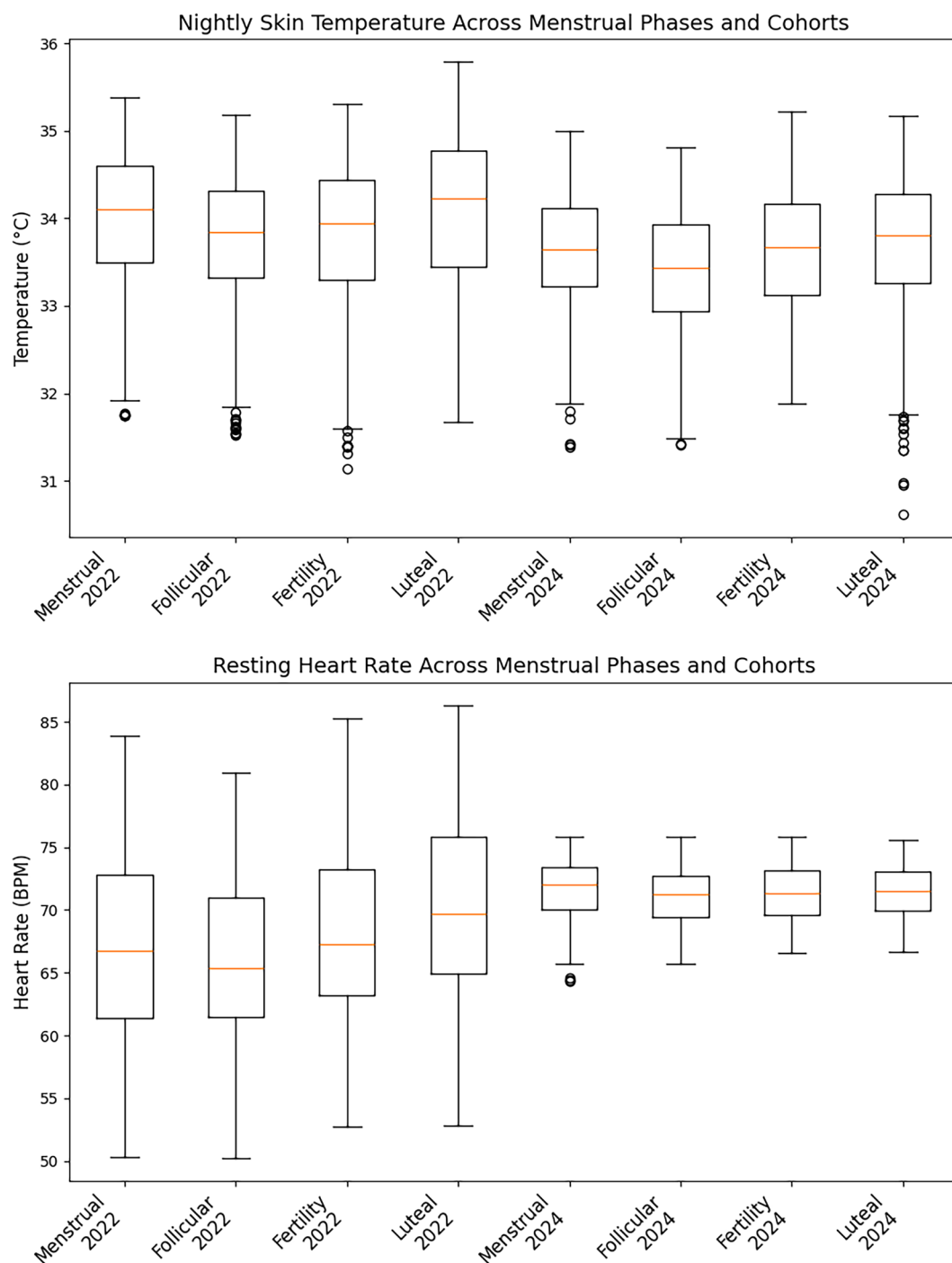


Fig. 2 Boxplots showing the distribution of (top) nightly wrist skin temperature and (bottom) resting heart rate within various phases of the menstrual cycle. The distributions are further split according to different subsets of our dataset (Interval 1, Interval 2, and both) to establish the persistence of these trends.

late-follicular), 44.4% (late-follicular to ovulation), 67.1% (ovulation to luteal), and 100% (luteal back to menstruation) according to dataset averages.

To support the validity of our dataset, Fig. 1 illustrates the smoothed average hormone levels across the menstrual cycle. The observed patterns closely aligned with established expectations from literature. LH levels exhibited a sharp mid-cycle peak, consistent with their role in triggering ovulation. This surge is known to result from rising estradiol levels, which exert positive feedback on the hypothalamic-pituitary axis, stimulating gonadotropin-releasing hormone (GnRH) release and subsequent LH secretion²⁵. Estrogen levels gradually increased and also peaked near the middle of participants' menstrual cycles, reflecting its role in initiating the LH surge via the hypothalamic-pituitary-gonadal axis²⁵. Meanwhile, progesterone rose during the second half of the cycle and peaked in the luteal phase, reflecting corpus luteum activity and the establishment of endometrial receptivity²⁶.

We also examined nightly wrist skin temperature and resting heart rate collected by the Fitbit, as these are two of the most commonly studied physiological markers in menstrual cycle research. More specifically, we investigated how these measures varied across menstrual cycle phases across both intervals of our study. Statistical analysis using repeated-measures ANOVA confirmed that median nightly skin temperature differed significantly across menstrual phases ($F(3, 123) = 11.63, p < .001$). As shown in Fig. 2, nightly temperature exhibited the cyclical pattern we anticipated; it was lowest during the follicular phase, rose during the fertile window, and peaked during the luteal phase before dropping again during menstruation. This pattern is consistent with prior findings that link elevated basal and core body temperatures to progesterone-driven thermogenic effects during the luteal phase^{15,27,28}.

Resting heart rate followed a similar phase-dependent pattern. Repeated-measures ANOVA also indicated significant phase-dependent variation in median resting heart rate ($F(3, 123) = 10.26, p < .001$). It was lowest during the follicular phase, increased through the fertile window, and peaked during the luteal phase. Previous studies suggest that this trend may be driven by increased sympathetic nervous system activity and elevated metabolic demands during the luteal phase^{29,30}. The subsequent return to baseline resting heart rate during menstruation aligns with the decline in estrogen and progesterone levels.

These consistent cyclical patterns replicated across two distinct data collection intervals support the internal validity of our dataset. The concordance of our findings with established hormonal-phase biomarkers strengthens confidence in the sensitivity and reliability of these wearable-derived signals for tracking menstrual cycle dynamics.

Data availability

The mcPHASES dataset and relevant documentation are accessible through PhysioNet (<https://physionet.org/content/mcphases/1.0.0/>).

Code availability

Example scripts demonstrating data import procedures and figure generation methodologies presented in this article are made available via the public repository at <https://github.com/chai-toronto/mcphases>.

Received: 7 October 2025; Accepted: 2 February 2026;

Published online: 10 February 2026

References

- Farage, M. A., Neill, S. & MacLean, A. B. Physiological changes associated with the menstrual cycle: a review. *Obstetrical & gynecological survey* **64**, 58–72 (2009).
- Thiyagarajan, D. K., Basit, H. & Jeanmonod, R. Physiology, menstrual cycle. In *StatPearls [Internet]* (StatPearls Publishing, 2024).
- Bull, J. R. *et al.* Real-world menstrual cycle characteristics of more than 600,000 menstrual cycles. *NPJ digital medicine* **2**, 83 (2019).
- RW, R. The normal menstrual cycle and the control of ovulation. *Principle and Practice of Endocrinology and Metabolism* 935–947 (2000).
- Vollmar, A. K. R., Mahalingaiah, S. & Jukic, A. M. The menstrual cycle as a vital sign: a comprehensive review. *F&S Reviews* 100081 (2024).
- American Academy of Pediatrics and Committee on Adolescence and American College of Obstetricians and Gynecologists and Committee on Adolescent Health Care. Menstruation in girls and adolescents: using the menstrual cycle as a vital sign. *Pediatrics* **118**, 2245–2250 (2006).
- Bobel, C. & Bobel, C. Beyond the managed body: putting menstrual literacy at the center. *The Managed Body: Developing Girls and Menstrual Health in the Global South* 281–321 (2019).
- Critchley, H. O. *et al.* Menstruation: science and society. *American journal of obstetrics and gynecology* **223**, 624–664 (2020).
- Holst, A. S. *et al.* Experiences of menstrual inequity and menstrual health among women and people who menstruate in the barcelona area (spain): a qualitative study. *Reproductive health* **19**, 45 (2022).
- National Academies of Sciences, Engineering, and Medicine. Overview of research gaps for selected conditions in women's health research at the national institutes of health: Proceedings of a workshop—in brief (2024).
- Goodale, B. M. *et al.* Wearable sensors reveal menses-driven changes in physiology and enable prediction of the fertile window: observational study. *Journal of medical Internet research* **21**, e13404 (2019).
- Jasinski, S. R., Presby, D. M., Grosicki, G. J., Capodilupo, E. R. & Lee, V. H. A novel method for quantifying fluctuations in wearable derived daily cardiovascular parameters across the menstrual cycle. *NPJ digital medicine* **7**, 373 (2024).
- Shilaih, M., Clerck, V. D., Falco, L., Kübler, F. & Leeners, B. Pulse rate measurement during sleep using wearable sensors, and its correlation with the menstrual cycle phases, a prospective observational study. *Scientific Reports* **7**, 1–7 (2017).
- Maijala, A., Kinnunen, H., Koskimäki, H., Jämsä, T. & Kangas, M. Nocturnal finger skin temperature in menstrual cycle tracking: ambulatory pilot study using a wearable oura ring. *BMC Women's Health* **19**, 1–10 (2019).
- Baker, F. C., Sibozza, F. & Fuller, A. Temperature regulation in women: Effects of the menstrual cycle. *Temperature* **7**, 226–262, <https://doi.org/10.1080/23328940.2020.1735927> (2020).
- Gloe, L. M., Block, S. R., Klump, K. L., Beltz, A. M. & Moser, J. S. Determining menstrual cycle phase: An empirical examination of methodologies and recommendations for improvement in behavioral and brain sciences. *Hormones and behavior* **155**, 105421 (2023).
- Yeung, E. H. *et al.* Longitudinal study of insulin resistance and sex hormones over the menstrual cycle: the biocycle study. *The Journal of Clinical Endocrinology & Metabolism* **95**, 5435–5442 (2010).
- Epstein, D. A. *et al.* Examining menstrual tracking to inform the design of personal informatics tools. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 6876–6888 (2017).
- Nakhuda, G. S., Li, N., Yang, Z. & Kang, S. At-home urine estrone-3-glucuronide quantification predicts oocyte retrieval outcomes comparably with serum estradiol. *F&S Reports* **4**, 43–48, <https://doi.org/10.1016/j.xfre.2023.01.006> (2023).
- Zhang, J. Algorithm effectiveness report of mira. <https://www.miracare.com/assets/algorithm-effectiveness.pdf> [Online; accessed 14-February-2023] (2022).
- Li, N. Precision performance of the mira confirm test and comparison with the semi-quantitatively product. <https://www.miracare.com/assets/precision-performance-of-the-mira-confirm-test-and-comparison-with-the-semi-quantitatively-product.pdf> [Online; accessed 14-February-2023] (2022).

22. Bouchard, T. P., Fehring, R. J. & Mu, Q. Quantitative versus qualitative estrogen and luteinizing hormone testing for personal fertility monitoring. *Expert review of molecular diagnostics* **21**, 1349–1360 (2021).
23. Goldberger, A. *et al.* Physiobank, physiotoolkit, and physionet: Components of a new research resource for complex physiologic signals. *Circulation* **101**, e215–e220 (2000).
24. Lin, B., Li, J. Y., Kalani, K., Truong, K. & Mariakakis, A. mcphases: A dataset of physiological, hormonal, and self-reported events and symptoms for menstrual health tracking with wearables, <https://doi.org/10.13026/zx6a-2c81> Version 1.0.0. RRID:SCR_007345 (2025).
25. Kauffman, A. S. Neuroendocrine mechanisms underlying estrogen positive feedback and the lh surge. *Frontiers in Neuroscience* **16**, 953252, <https://doi.org/10.3389/fnins.2022.953252> (2022).
26. Abdullah, S. *et al.* Distinct urinary progesterone metabolite profiles during the luteal phase. *Hormone Molecular Biology and Clinical Investigation* **44**, 137–144, <https://doi.org/10.1515/hmbci-2022-0065> (2022).
27. Buxton, C. L. & Atkinson, W. B. Hormonal factors involved in the regulation of basal body temperature during the menstrual cycle and pregnancy. *The Journal of Clinical Endocrinology* **8**, 544–549 (1948).
28. Lin, G. *et al.* Understanding wrist skin temperature changes to hormone variations across the menstrual cycle. *npj Women's Health* **2**, 35 (2024).
29. Brar, T. K., Singh, K. & Kumar, A. Effect of different phases of menstrual cycle on heart rate variability (hrv). *Journal of Clinical and Diagnostic Research* <https://doi.org/10.7860/jcdr/2015/13795.6592> (2015).
30. Leicht, A. S., Hirning, D. A. & Allen, G. D. Heart rate variability and endogenous sex hormones during the menstrual cycle in young women. *Experimental physiology* **88**, 441–446 (2003).

Acknowledgements

This work was supported in part by NSERC Discovery Grants RGPIN-2021-03457 and RGPIN-2021-04268, a Google PhD Fellowship, and an unrestricted research gift from Google.

Author contributions

G.L., K.T., and A.M. conceived the studies. G.L. conducted the studies. G.L., J.L., and K.K. analyzed the data. All authors reviewed the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to G.L.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.



Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2026